1. Light is a **self-propagating electromagnetic disturbance**: an $\vec{E}$ field and a $\vec{B}$ field, perpendicular to each other and to the direction of travel.

A. In the box below, first draw a **right-handed** set of coordinate axes and label x, y, and z. Then sketch **one full wavelength** of a light wave traveling in the $+x$ direction whose $\vec{E}$ field oscillates along $\pm y$. Label $\vec{E}$ and $\vec{B}$, and make sure the $\vec{B}$ you draw is consistent with the direction of travel.

B. At a place where your $\vec{E}$ points along $+y$, which way does $\vec{B}$ point? _____

C. Light travels in the direction of the Poynting vector, $\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$. Use it to fill in the direction of travel:

$\vec{E}$ along $+y$, $\vec{B}$ along $+z$ $\implies$ travels _____

$\vec{E}$ along $+z$, $\vec{B}$ along $+x$ $\implies$ travels _____

$\vec{E}$ along $-y$, $\vec{B}$ along $+z$ $\implies$ travels _____

D. Now go backwards. A wave travels in the $+z$ direction and its $\vec{E}$ field is along $+x$. Which way is $\vec{B}$? _____

2. A light wave travels in the $+x$ direction. Its electric field oscillates along $\pm y$ with a peak value of $E_0 = 850$ N/C, and

$$k = 1.20 \times 10^7 \text{ rad/m} \qquad \omega = 3.60 \times 10^{15} \text{ rad/s}$$

A. Write the full traveling wave equation for the electric field, $E(x, t)$.

B. Show that this wave really does travel at the speed of light.

C. What is the peak magnetic field, B_0?

D. Write the full traveling wave equation for the magnetic field, $B(x, t)$. Which axis does $\vec{B}$ lie along?

E. Find the wavelength λ and the frequency f. Is this light something your eye could see?